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Flexible Data Aggregation for Prediction and Decision Making with Contextual Information: Applications in Retailing

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Abstract. ***Problem definition*:** How should online retailers make demand predictions or operational decisions with limited relevant data? Motivated by conventional data aggregation approaches, we develop a flexible data aggregation (FlexDA) framework to adapt to different degrees of heterogeneity across products, thereby striking a better balance between the bias and variance of data-driven predictions and decisions. ***Methodology/results*:** Under the FlexDA framework, we propose aggregating the individual data sets at different levels, potentially based on contextual information, and training submodels with each aggregated data set. A meta-model is trained to integrate the outputs of these submodels with a set of weights. For demand prediction tasks with linear models, we propose a consistent estimate of the optimal weight and theoretically demonstrate the advantage of the FlexDA approach over existing approaches under the small-data large-scale regime. For decision making with nonlinear feature-demand relationships, with a fixed sample size, we show that the optimality gap of the FlexDA approach decays near-linearly in the number of products with high probability. We further validate the FlexDA approach with synthetic data and real data from the Rossmann store. Building on theoretical development and empirical validation, we conducted an internal study with Meituan, focusing on ordering problems in their community group buying business. Our proposed approach achieves an average reduction of more than 10% in both lost sales ratio and inventory ratio for newly launched fresh products, as well as standard products compared with the algorithm implemented by Meituan. ***Managerial implications*:** Simply aggregating data from all products and training a shared model reduces the high variance caused by data scarcity but compromises the ability to capture heterogeneity across products. Our study highlights the value of flexible data aggregation for data-driven prediction and decision making, especially for large-scale applications with limited data, with both theoretical and empirical support.

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1. Introduction

Fresh foods have long been an advantage for brick-and-mortar stores over online e-commerce giants due to consumers' preference for freshness (Supermarket News 2019). However, demand for fresh foods at individual brick-and-mortar stores can be volatile, leading to substantial demand and supply mismatches, which in turn result in food spoilage and waste. On the other hand, traditional online food sales, aiming to expand the market base by offering fresh products, face tremendous challenges in maintaining cold-chain logistics when shipping products from central warehouses to individual households. To address these challenges, innovative business models, such as community group buying (CGB), have emerged in China, such as Duoduo Grocery and Meituan Grocery Retailing (formerly known as Meituan Youxuan; hereafter referred to as "Meituan" for brevity). In this business model, online platforms partner with community leaders who attract local consumers to place orders online and serve as neighborhood pickup points. With innovations in operational modes, China's CGB market grew from US$1.3 billion in 2018 to US$45.6 billion in 2023, a remarkable 3407% increase within five years (ECDB 2024).

Our industry collaborator, Meituan, a leading Chinese retail platform for local consumer products and

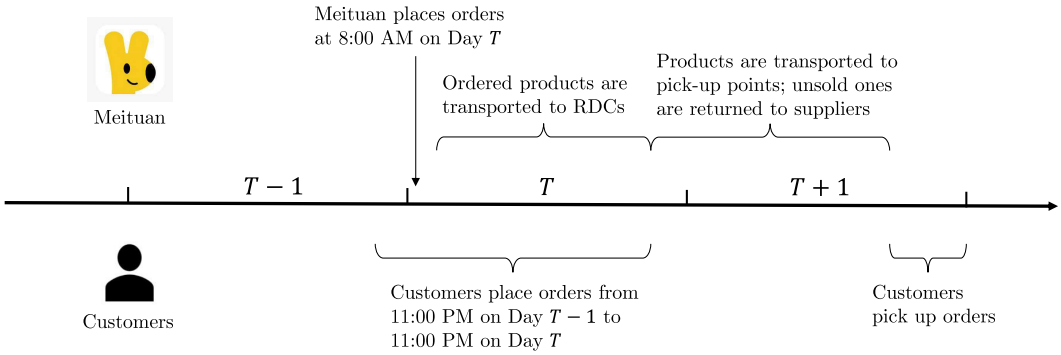
retail services, has become the largest player in China's fast-growing CGB market. Its success largely rests on the prompt and efficient delivery of fresh products and the wide variety of product offerings. In particular, as illustrated in Figure 1, its procurement plans should be prescribed daily based on limited order information to balance the freshness and timely delivery of ordered products. Furthermore, it offers a limited daily assortment to consumers with frequent rotations. For instance, Figure 2 shows that the fish assortment was changed overnight on two consecutive days in January 2024. The benefits of rotating selective offerings are twofold: (i) demand becomes more concentrated, enhancing economies of scale for each stock keeping unit (SKU), and (ii) they encourage variety-seeking consumers to visit Meituan's app regularly and engage in discovering new offerings.

Although Meituan's innovative operational mode secures its leading CGB market share, it also gives rise to several operational challenges. The short time window between consumers' order placements and Meituan's subsequent orders to suppliers requires accurate demand predictions and effective procurement strategies. Because procurement cannot be fully make-to-order, certain customer orders may remain unfulfilled, leading to lost sales, whereas unsold items at the regional distribution centers (RDCs) are returned to suppliers. Furthermore, frequent product rotation not only limits the availability of historical data for each product but also inevitably increases the number of products to be managed concurrently. Thus, leveraging limited demand observations to facilitate operational decision making in such a large-scale system is both a challenging task and a critical driver to Meituan's success.

Although data have become increasingly important in shaping operational and marketing decisions in retailing, such as demand prediction (Cui et al. 2015, Guo et al. 2022), product pricing (Zhao et al. 2025), and inventory planning (Huang et al. 2025), the data scarcity issue faced by Meituan is not an exception. This is due to the volatile economic environment and rapid launch of new products, which frequently reshape demand patterns and result in the scarcity of relevant data (Feng et al. 2023). Moreover, innovative retail operations, such as frequent assortment rotation at Meituan, may further reduce data availability. In addition, with the advancement of information technology, retailers nowadays collect not only historical sales data but also contextual information such as product features, consumer behaviors, weather forecasts, economic factors, and so on (Chu et al. 2025). For example, Meituan records more than 300 features in its sales data. The high-dimensional covariate information, which is potentially valuable, may further intensify the extent of data sparsity. Although machine learning–based models have been developed for demand prediction and inventory replenishment and have demonstrated advantages in data-rich environments (Ban and Rudin 2019, Qi et al. 2020, Makridakis et al. 2022), their applicability to environments with limited data, such as Meituan, remains unclear.

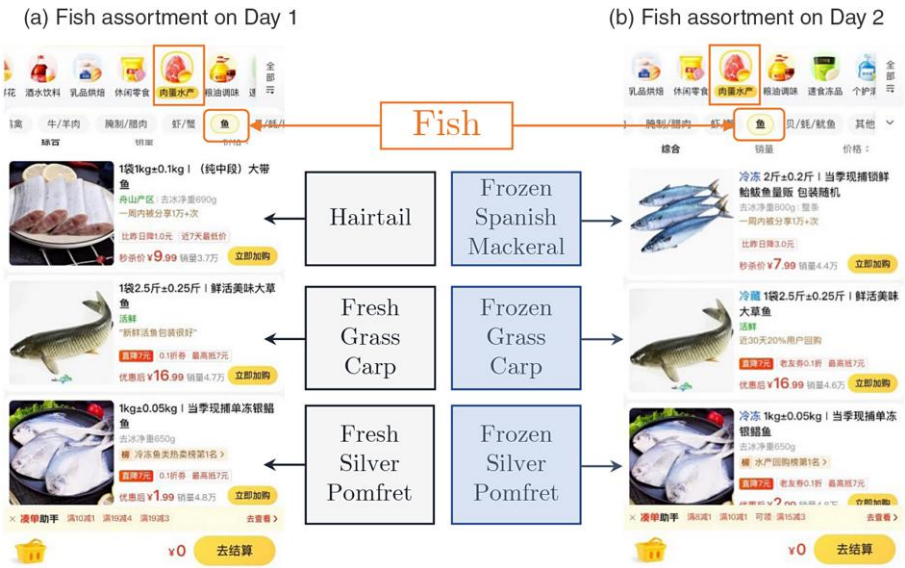
Demand prediction and decision making with limited data for thousands of products are challenging for retailers, known as the small-data large-scale regime (Gupta and Rusmevichientong 2021). To tackle this issue, Meituan's supply chain algorithm team employed a *full-aggregation* strategy. They pooled available data across all products as a single data set, which can effectively address the issue of data scarcity. A shared model is then trained on this fully aggregated data set, and the demand predictions or order quantity decisions are made based on this shared model for all products. This full-aggregation strategy has demonstrated its superior performance due to its ability to reduce variance with an increased data size (Chuang et al. 2021, Makridakis

Figure 1. (Color online) Timeline of the Community Group Buying Business at Meituan



Notes. (1) Customers place orders between 11:00 p.m. on day $T-1$ and 11:00 p.m. on day T and collect the goods from designated pickup point on day $T+1$. (2) At 8:00 a.m. on day T, Meituan sends the procurement plan to suppliers to ensure that goods are delivered to the regional distribution centers (RDCs) on the same day. (3) On day $T+1$, the goods are transported to the frontier distribution centers (FDCs) and subsequently dispatched to the designated pickup points.

Figure 2. (Color online) Fish Assortments Offered on Two Consecutive Days



Note. Screenshots of the fish assortments were retrieved on January 3 and 4, 2024.

et al. 2022). Although the implementation of this approach is not uncommon in practice, especially when products are similar, it often faces criticism (Lei et al. 2025). In particular, simply pooling all data sets overlooks the heterogeneity across products, that is, the variability of the impacts of features on the products. Furthermore, it is silent on how to leverage data sets aggregated at different levels.

On the other hand, one may also adopt the decoupled approach that only utilizes individual data sets to effectively capture the heterogeneity across products to reduce bias. This *no-aggregation* strategy, an extreme case from the perspective of data aggregation, is known to be able to recover the optimal solutions with a sufficient amount of data; for example, implementing sample average approximation (SAA) for each product (Shapiro et al. 2021). However, it may perform poorly with limited data (Besbes and Mouchtaki 2023). Therefore, existing studies strive to improve the decoupled approach via various techniques, such as regularization (Ban and Rudin 2019), shrinkage (Gupta and Kallus 2022), and transfer learning (Bastani 2021).

In summary, existing approaches to demand prediction and decision making under limited data face an inherent tradeoff: Full aggregation reduces variance by pooling data but ignores product heterogeneity, whereas no aggregation preserves heterogeneity but suffers from high variance under data scarcity. To improve the bias–variance tradeoff under the small-data large-scale regime, inspired by characteristics of Meituan's business model, we propose the flexible data aggregation (FlexDA) framework for demand prediction and decision making with contextual information. In particular, the FlexDA approach allows for data aggregation at different levels, potentially by incorporating domain knowledge. A series of submodels, potentially from different model classes, are trained with different aggregated data sets to capture different degrees of heterogeneity across products. Their outputs are then combined through a meta-model that adaptively assigns data-driven weights, thereby carefully balancing bias and variance. Rather than relying on a fixed model class built on a single aggregation scheme, our framework is flexible in leveraging a mixture of submodels trained under diverse aggregation schemes, which gives it a competitive edge in small-data large-scale regimes.

We first investigate the FlexDA method under linear models with two-level data aggregation for demand prediction. We characterize the optimal weights assigned to each submodel, which guarantees the advantage of the proposed approach over conventional aggregation approaches. We then propose a consistent estimator of the optimal weight in a data-driven setting, ensuring the framework's practical applicability and effectiveness. We extend our analysis to the implementation of the FlexDA method for decision making, which can incorporate nonlinear feature-response relationships and general cost functions. We show that, with a fixed sample size, the optimality gap of the FlexDA approach decays near-linearly (up to a logarithmic factor) in the number of products with high probability. Such a performance guarantee justifies the applicability of the FlexDA approach for Meituan's ordering problem, as well as in various operational settings. For both prediction and decision making, we demonstrate the superior performance and

robustness of FlexDA methods over existing ones with synthetic and real data. Building on the theoretical development and empirical validation, we conducted an internal study with our industry collaborator, Meituan, by building a customized FlexDA model for the ordering problem for their CGB platform. The numerical results show that our FlexDA approach outperforms the existing state-of-the-art approaches. This underscores the significance of flexible data aggregation for prediction and decision making under the small-data large-scale regime.

Our contributions are threefold, in terms of modeling, analysis, and practical implementation.

1. **Data aggregation with contextual information.** We develop a FlexDA framework that can aggregate individual data sets, potentially using contextual information, to adapt to different degrees of heterogeneity across multiple products with different submodels, which are combined by a set of tunable weights. This framework not only generalizes the conventional approaches but also can accommodate multiple levels of data aggregations, nonlinear decision rules, and general objective functions, guaranteeing a wide range of its applications in practice.

2. **Analysis and performance guarantee.** We analyze the FlexDA method under linear models for demand prediction and show that as the number of products increases, the data-driven weights combining the submodels converge to the optimal ones. Our theoretical results demonstrate the benefit of the FlexDA method over the existing approaches, which indicates the potential superior performance of the FlexDA method in practice under the small-data large-scale regime. Furthermore, we provide a performance guarantee of the generalized FlexDA approach with nonlinear models and general costs. We show that the optimality gap of the FlexDA approach decays near-linearly (up to a logarithmic factor) in the number of products with high probability. This justifies the applicability of the proposed FlexDA approach in various operational settings.

3. **Industry collaboration with Meituan.** We conducted an internal study with Meituan to solve their ordering problem for the CGB business. The FlexDA method demonstrated significant performance improvements over the existing approach adopted by the platform and other state-of-the-art machine learning models. For fresh products, the FlexDA method reduced the inventory ratio (IR) by 42% and lowered the lost sales ratio (LSR) by 14% compared with Meituan's existing approach (the IR and LSR are the key performance metrics adopted by Meituan, which are formally defined in Equation (24) in Section 7). For standard products, the model achieved a more than 10% reduction in both IR and LSR. These results underscore the effectiveness of the FlexDA approach in optimizing order quantity in the retailing industry and reducing lost sales across various product categories.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. In Section 3, we introduce the problem setup. In Section 4, we formally present the FlexDA framework. Section 5 investigates the FlexDA method under linear models for prediction tasks. In Section 6, we investigate the performance of the FlexDA approach for decision making with nonlinear models and convex objectives. Section 7 provides the details of implementing the FlexDA method at Meituan and reports the key performance metrics compared with the existing approach. Section 8 concludes the paper.

2. Related Literature

Our work relates to the stream of literature on demand prediction and retail operations and to the growing body of research on small-data large-scale regimes.

2.1. Demand Prediction and Retail Operations

Demand prediction has been playing a pivotal role in retail management, supplying key inputs to the optimization problem for formulating operational and retailing decisions. A significant body of literature has been devoted to improving demand prediction performance in retailing settings (Cohen et al. 2022b, Fildes et al. 2022). Although one group of studies investigates the value of incorporating contextual information for demand prediction (Kesavan et al. 2010, Cui et al. 2018, Fildes et al. 2019, Huang et al. 2019), another group focuses on developing sophisticated regression and machine learning–based models for enhancing predictive power (Ma et al. 2016; Guo et al. 2022, 2025; Makridakis et al. 2022; Arora et al. 2023).

Because of the high variability in SKU-level demand data or data scarcity during model calibration, data aggregation has been commonly adopted to improve demand prediction (Syntetos et al. 2016). Cohen et al. (2022a) proposed the data aggregation and clustering (DAC) algorithm and investigated the benefit of data aggregation with generalized linear models. Lei et al. (2025) developed the pooling and boosting (PAB) method for retail demand forecasting. Relying on the assumption that all products share the same model parameters to materialize the benefit of data pooling, the performance of the PAB method can be less satisfactory when the degree of heterogeneity among products is significant. Chen and Zhao (2025) study how to aggregate experts' forecasts across multiple variables, showing that pooling forecasts can improve consensus accuracy compared with aggregating forecasts for each variable separately. In contrast to existing data aggregation approaches, our method not only allows flexible data aggregation, potentially by integrating domain

knowledge, but also carefully integrates submodels obtained at different aggregation levels via a metamodel.

Instead of executing an estimate-then-optimize procedure for decision-making in retail operations, many researchers seek an end-to-end framework that directly translates observable data into operational decisions (Liyanage and Shanthikumar 2005, Elmachtoub and Grigas 2022). The data-driven newsvendor problem is a fundamental one that has been extensively explored in the literature (Feng and Shanthikumar 2022). Liyanage and Shanthikumar (2005) and Chu et al. (2008) introduce the notion of operational statistics that integrates parameter estimation and profit optimization for the newsvendor problem. Feng and Shanthikumar (2023) propose the general operational data analytics (ODA) framework, which emphasizes identifying an appropriate class of operational statistics and unifies many existing data-driven approaches under this framework. Although our model aligns with this perspective, our contribution is to enhance model diversity by leveraging various data aggregation levels and to characterize the weights that combine these submodels in order to improve the bias–variance tradeoff across multiple data-driven decisions. A stream of studies focuses on the efficiency of solving the SAA counterpart of the problem (Levi et al. 2007, 2015; Besbes and Mouchtaki 2023). More recently, a growing number of studies have adopted machine learning models to solve the data-driven newsvendor problem with contextual information. Ban and Rudin (2019) study the newsvendor problem with the features via the empirical risk minimization (ERM) and the kernel methods. Oroojlooyjadid et al. (2020) adopted deep neural networks to solve the newsvendor problem. Our collaboration with Meituan is related to existing studies on the data-driven newsvendor problem that typically focus on big data regimes. Our work aims to propose an effective data-driven approach for solving the ordering problems of a large number of products with limited data. Furthermore, to align with the company's operational objectives in practice, instead of adopting the newsvendor loss directly, we tailor the training objective function to balance two key performance metrics, as described in detail in Section 7, which increases analytical complexity.

Another relevant stream of research focuses on causal inference approaches for demand estimation, including experimental methods (Fisher et al. 2018), instrumental variables (Jiang and Li 2025), and structural estimation (Adam et al. 2024). These studies primarily aim to obtain unbiased estimates of the (average) treatment effect, which is essential when the objective is to optimize pricing decisions. In contrast, our collaboration with Meituan centers on demand prediction and inventory management, where the goal is to minimize the mean squared error of predictions or reduce inventory-related costs. In this context, it is often preferable to accept some bias in order to reduce variance, because this tradeoff leads to improved overall performance in operational decision making.

2.2. Small-Data Large-Scale Regime

The goal of our work is to facilitate Meituan to solve the ordering problems for a large number of products with limited data. This corresponding model setup is defined as the small-data large-scale regime by Gupta and Rusmevichientong (2021). The pioneering work by Stein (1956) introduces the idea of data pooling for the simultaneous estimation of multiple Gaussian means and demonstrates its benefit over the decoupled approach, a result known as Stein's phenomenon. Many follow-up works aim to explain Stein's phenomenon (Brown 1971, Efron and Morris 1977, Stigler 1990, Beran 1996, Brown and Zhao 2012). Gupta and Kallus (2022) extend Stein's method to data-driven optimization problems and develop the shrunken-SAA algorithm. Their method improves the decoupled sample average approximation approach by shrinking the individual-level data to an anchor distribution. Furthermore, Li et al. (2023) leverage clustering techniques to improve the performance of data pooling. Recently, many studies have emphasized addressing the challenge of limited data in a dynamic system (Goh et al. 2018, Zhang et al. 2025). Our proposed FlexDA approach is related to the idea of shrinkage, but it does not necessarily hinge on a single anchor distribution. In particular, our FlexDA approach can combine submodels derived from data sets aggregated at different levels. Moreover, our approach also incorporates contextual information, which is a key input to modeling and analyzing current retail operations.

Our work is also related to transfer learning, which improves a learner from one domain by transferring information from a related domain (Harel and Mannor 2011, Wang and Mahadevan 2011, Duan et al. 2012, Zhou et al. 2014, Nam and Kim 2015). Bastani (2021) proposes a two-step estimator to efficiently combine a large amount of proxy data and a small amount of target data. Bastani et al. (2022) apply transfer learning to the problem of learning shared structure via a sequence of dynamic pricing experiments for related products. Nabi et al. (2022) propose a hierarchical empirical Bayes approach to learn empirical meta-priors. Feng et al. (2023) develop cross-learning solutions under the parametric ODA framework. Different from the aforementioned studies based on transfer learning, our work focuses on solving multiple problems simultaneously with limited data for each one, without knowing the parametric distributions of the random variables.

3. Problem Setup

To ensure the timely delivery of fresh products and capture the volatile demand patterns during our collaboration with Meituan, a single-period inventory replenishment problem is solved on a daily basis. In particular, the information available up to 8:00 a.m. is used to determine the order quantities for each day. This contextual information includes category details, selling prices, consumer clicks and purchased amounts recorded before 8:00 a.m., and other relevant features. In total, the number of features exceeds 300. On the other hand, because of the frequent rotation of product offerings, the effective historical demand data for the products listed in Meituan's app are rather limited. This requires the development of a data-driven decision-making framework for managing a large-scale inventory system with limited data.

To solve Meituan's replenishment problem, we consider a firm managing multiple products, denoted by $[K] = \{1, 2, \ldots, K\}$, with random demands associated with contextual information. In particular, for each product k, the random demand D_k is related to a feature vector, $\boldsymbol{X}_k = (X_{k,1}, X_{k,2}, \ldots, X_{k,F}) \in \mathbb{R}^F$, where F is the number of features. The firm solves the following contextual stochastic optimization problem:

$$\min_{q_k \in \mathcal{Q}_k, k \in [K]} \frac{1}{K} \sum_{k \in [K]} \mathbb{E}_{D_k}[c_k(q_k, D_k) | \boldsymbol{X}_k = \boldsymbol{x}_k], \tag{1}$$

where the expectation is taken with respect to the conditional distribution of the random demand D_k given the newly observed feature vector $\boldsymbol{x}_k = (x_{k,1}, x_{k,2}, \ldots, x_{k,F})$, and q_k is the decision for product k, which should be constrained within the feasible region $\mathcal{Q}_k$. The cost function for product k, $c_k(q_k, d_k)$, depends on the decision variable q_k and the realized demand d_k. For example, if c_k takes the form of mean square error (MSE), that is, $c_k(q_k, d_k) = (q_k - d_k)^2$, this implies that the firm focuses on estimating the demand means of all products. If c_k is in the form of newsvendor loss, that is, $c_k(q_k, d_k) = c_b(d_k - q_k)^+ + c_h(q_k - d_k)^+$, where c_b and c_h are the per-unit back-order and holding costs, this represents the case where the firm determines the order quantities for all products.

Without the full characterization of the conditional distribution of random demands, the only accessible information is the historical data set, which contains the past contextual information and the realized demands. For product k, the set of observed feature-demand pairs is denoted by $\mathcal{S}_k = \{(\hat{\boldsymbol{x}}_k^i, \hat{d}_k^i) : i \in [N_k]\}$, where $[N_k] = \{1, 2, \ldots, N_k\}$ is the set of indices of historical observations for product k, and $\hat{\boldsymbol{x}}_k^i$ and $\hat{d}_k^i$ denote the observed feature vector and realized demand for the ith sample, respectively. For ease of exposition, let $\hat{X}_k = [\hat{\boldsymbol{x}}_k^1, \hat{\boldsymbol{x}}_k^2, \ldots, \hat{\boldsymbol{x}}_k^{N_k}]^\top \in \mathbb{R}^{N_k \times F}$ denote the observed feature matrix and $\hat{D}_k = [\hat{d}_k^1, \hat{d}_k^2, \ldots, \hat{d}_k^{N_k}]^\top \in \mathbb{R}^{N_k}$ the observed demand vector for product k. Given the whole data set $\mathcal{S} = \{\mathcal{S}_k, k \in [K]\}$, the objective of the firm is to identify a set of functions $\{h_k \in \mathcal{H}_k, k \in [K]\}$ from $\mathcal{H}_k = \{h : \mathbb{R}^F \to \mathcal{Q}_k\}$, $k \in [K]$, the classes of functions mapping the feature space into the decision space for product k. For example, $\mathcal{H}_k$ contains the class of linear functions that translate contextual features into final decisions. Alternatively, $\mathcal{H}_k$ may include the classes of nonlinear models, such as nonlinear regression or machine learning models, that characterize the nonlinear relationships between features and the final outputs.

Given the additive objectives in Equation (1) and the independence across random demands, a canonical approach, that is, no aggregation, is to solve each problem separately based on the respective historical data:

$$\hat{h}_k(\mathcal{S}_k) \in \operatorname*{argmin}_{h_k \in \mathcal{H}_k} \frac{1}{N_k} \sum_{i=1}^{N_k} c_k(h_k(\hat{\boldsymbol{x}}_k^i), \hat{d}_k^i), \; k \in [K]. \tag{2}$$

This decoupled approach, which adopts the ERM for each product, can recover the optimal solutions with a sufficient amount of data (Shapiro et al. 2021), but may perform poorly due to the large variance with limited data (Besbes and Mouchtaki 2023). To address the issue of data scarcity, an alternative approach, that is, full aggregation, is to aggregate data from all products and approximate the distribution of the random demand using the empirical distribution of the aggregated data:

$$\hat{h}_0(\mathcal{S}) \in \operatorname*{argmin}_{h_0 \in \mathcal{H}_0} \frac{1}{\mathbb{N}} \sum_{l=1}^{K} \sum_{i=1}^{N_l} c_k(h_0(\hat{\boldsymbol{x}}_l^i), \hat{d}_l^i), \; k \in [K], \tag{3}$$

where $\mathbb{N} = \sum_{k=1}^{K} N_k$ denote the total sample size. Although the second approach may mitigate the high variance of the solutions by utilizing ample data (Chuang et al. 2021), it neglects the heterogeneity among products, which could lead to significant bias in the solutions for each product (Lei et al. 2025).

Although the first approach, relying solely on individual data sets, can help capture the product-specific feature-response relationships across products, the second one utilizes fully aggregated data, aiming to reduce the variance of predictions and decisions, and can provably enhance performance when products are similar. In light of these two extreme aggregation strategies, one may flexibly calibrate the level of data aggregation to capture different degrees of heterogeneity across products. In the next section, we formally introduce the FlexDA framework for prediction and decision making in a data-driven setting. In a nutshell, the FlexDA approach first trains multiple submodels with different levels of data aggregation to capture different degrees of heterogeneity across products and then aggregates those submodels into the final output using a set of tunable weights.

4. The Flexible Data Aggregation (FlexDA) Framework

In this section, we formally introduce the FlexDA framework, inspired by Meituan's operational characteristics. Each region, served by a dedicated RDC, launches thousands of products, which can be grouped into four level 1 categories, as shown in Figure 3, including "Fresh Food," "Alcoholic Beverages and Snacks," "Grain and Daily Necessities," and "Others." Although the CGB business model primarily focuses on fresh goods, Meituan also offers products in the other three categories, collectively referred to as "Standard Products." Furthermore, within each level 1 category, Meituan further divides the products into several segments referred to as level 2 categories. For example, "Fresh Food" contains "Meat," "Vegetables," and so forth. In total, there are 14 level 2 categories. Products that share the same category at different levels often exhibit a certain degree of homogeneity; that is, the impacts of contextual information are similar across them. This motivates the design of the FlexDA framework, which can leverage product hierarchy information, as well as other contextual product information, to enhance the effectiveness of data aggregation.

4.1. Description of FlexDA Algorithm

To formally present the FlexDA framework, we first introduce the definition of *partition* of a data set that represents how the entire data set $\mathcal{S} = \{\mathcal{S}_k, k \in [K]\}$ is aggregated.

Definition 1. A set of data sets $\mathcal{P} = \{C_i, i \in [I]\}, I \in \mathbb{Z}^+$, is a partition of a data set $\mathcal{S} = \{\mathcal{S}_k, k \in [K]\}$ if the following conditions hold

(i) $C_i \cap \mathcal{S} \neq \emptyset$ for $i \in [I]$. In addition, either $C_i \cap \mathcal{S}_k = \mathcal{S}_k$ or $C_i \cap \mathcal{S}_k = \emptyset$ for $k \in [K]$;

(ii) $\cup_{i \in [I]} C_i = \mathcal{S}$;

(iii) $C_i \cap C_j = \emptyset$, $\forall\, i, j \in [I]$ and $i \neq j$.

The above definition of partition is a reminiscence of the partition of a set in the context of grouping data sets into nonempty data sets. Condition (i) implies that each set of the partition $\mathcal{P}$ can only contain complete individual data sets $\mathcal{S}_k$, $k \in [K]$. Condition (ii) indicates that the sets in the partition $\mathcal{P}$ cover the data set $\mathcal{S}$. Condition (iii) requires that sets in the partition $\mathcal{P}$ are mutually exclusive.

We illustrate possible partitions of the data set $\mathcal{S}$ with the following three cases.

- *No aggregation*: In Section 3, we discuss the decoupled ERM approach that treats each problem separately. Thus, the associated partition groups the data samples according to the products' indices, that is, $\mathcal{P} = \{\mathcal{S}_k, k \in [K]\}$.
- *Full aggregation*: In Section 3, we present the full-aggregation approach that pools all data sets into a single one. Thus, the partition contains a single set, which is the fully aggregated data set, $\mathcal{P} = \{\mathcal{S}\}$.
- *Context-based partition*: Suppose some feature X_l is a categorical variable that takes a finite number of possible values, such as sales regions and product categories, denoted by $\{a_1, a_2, \ldots, a_L\}$, and is fixed for each product (e.g., an inherent or time-invariant characteristic). One may group data sets by the value of X_l. The feature-based partition implied by feature l is given by $\mathcal{P} = \{\{(\hat{\boldsymbol{x}}_k^i, \hat{d}_k^i) \in \mathcal{S}_k : \hat{\boldsymbol{x}}_{k,l}^i = a_j, k \in [K]\}, j \in [L]\}$.

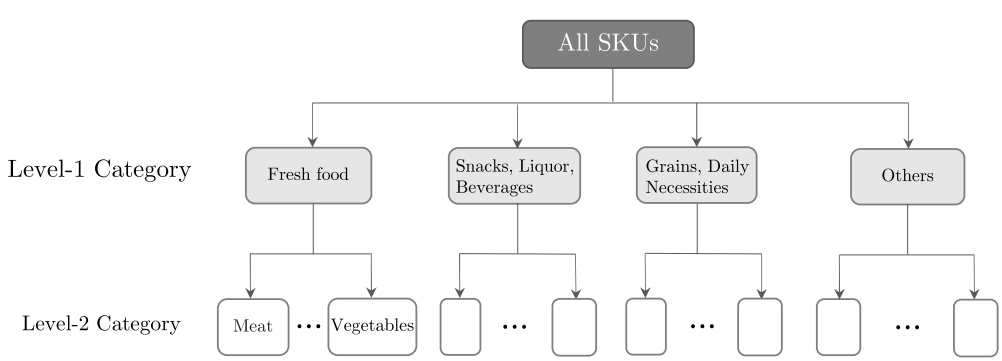
The FlexDA framework allows for aggregating (equivalently, partitioning) data samples flexibly, which are not necessarily restricted to the above three cases, to train submodels that capture the different degrees of heterogeneity among products. We next demonstrate the FlexDA approach with M partitions of the data set $\mathcal{S}$, denoted by $\{\mathcal{P}_m, m \in [M]\}$.

Step 1: For each partition $\mathcal{P}_m$, which contains l_m aggregated data sets, we fit submodels from the class of functions $\mathcal{H}_m$, respectively, using each aggregated data set: $(\mathcal{P}_m)_l$, $l = 1, 2, \ldots, l_m$. Specifically, the submodel trained on the l-th aggregated data set of partition $\mathcal{P}_m$ is obtained as follows:

$$\hat{h}_{m,l} = \arg\min \left\{ h \in \mathcal{H}_m : \sum_{\mathcal{S}_k \in (\mathcal{P}_m)_l} \sum_{(\boldsymbol{x}, d) \in \mathcal{S}_k} c_k(h(\boldsymbol{x}), d) \right\}. \quad (4)$$

Step 2: For each product $k \in [K]$, based on the newly observed feature vector, $\boldsymbol{x}_k$, we compute the outputs of

Figure 3. Product Hierarchy at Meituan



the submodels that are trained with the aggregated data sets, which contain the data set $\mathcal{S}_k$. Because of the partition's definition, there are M outputs, each corresponding to a specific partition. Then, those outputs are combined with a set of weights, $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_M)$, where α_m denotes the weight associated with partition m, and satisfy $\sum_{m=1}^{M} \alpha_m = 1$. The final output for product k given the contextual information $\boldsymbol{x}_k$ is

$$\hat{h}_k(\boldsymbol{x}_k) = \sum_{m=1}^{M} \alpha_m \cdot \left(\sum_{l=1}^{l_m} \hat{h}_{m,l}(\boldsymbol{x}_k) \mathbb{1}_{\{\mathcal{S}_k \in (\mathcal{P}_m)_l\}} \right). \tag{5}$$

One key component of the FlexDA method is the weight assigned to each submodel. When predicting demands with linear models, one can derive the closed-form optimal weights if the true distribution information were known, which suggests the form of a data-driven estimator (see Section 5). However, in general, with a nonlinear feature-demand relationship and general cost functions, it is challenging to analytically determine the value of weights. Therefore, we turn to cross-validation to learn these weights. Algorithm 1 formally outlines the general procedure to generate predictions or decisions via flexible data aggregation, which involves multiple partitions, nonlinear models, and general cost functions.

Algorithm 1 (FlexDA for Prediction and Decision Making)

Input: Data sets $\mathcal{S} = \{\mathcal{S}_k, k \in [K]\}$, a set of partitions $\{\mathcal{P}_m, m \in [M]\}$, model classes $\{\mathcal{H}_m, m \in [M]\}$, loss functions $\{c_k(\cdot), k \in [K]\}$, the number of cross-validation folds J, and a grid $\mathcal{A} = \{(\alpha_1, \ldots, \alpha_M) : \alpha_m \in [0,1)$ and $\sum_{m \in [M]} \alpha_m = 1\}$.

for $m = 1, 2, \ldots, M$ **do**
  for $l = 1, 2, \ldots, l_m$ **do**
    $\hat{h}_{m,l} = \arg\min\{h \in \mathcal{H}_m : \sum_{(\boldsymbol{x},d) \in (\mathcal{P}_m)_l} c_k(h(\boldsymbol{x}), d)\}$ // Train submodels
  end for
end for
for $k = 1, 2, \ldots, K$ **do**
  $\{\mathcal{S}_{k,1}, \ldots, \mathcal{S}_{k,J}\} \leftarrow \mathcal{S}_k$ // Divide the data set $\mathcal{S}_k$ into J folds with equal size
  for $j = 1, 2, \ldots, J$ **do**
    $\mathcal{S}_{k,-j} \leftarrow \mathcal{S}_k \setminus \mathcal{S}_{k,j}$ // Remove the data set $\mathcal{S}_{k,j}$ from $\mathcal{S}_k$ and reserve it for validation
  end for
end for
for $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_M) \in \mathcal{A}$ **do**
  for $j \in 1, \ldots, J$ **do**
    for $m = 1, 2, \ldots, M$ **do**
      $\mathcal{P}_{m,-j} = (C_{1,-j}, C_{2,-j}, \ldots, C_{l_m,-j}) \leftarrow \mathcal{S}_{-j} = \{\mathcal{S}_{k,-j}, k \in [K]\}$ // Aggregate $\mathcal{S}_{-j}$ by partition $\mathcal{P}_m$
      for $l = 1, 2, \ldots, l_m$ **do**
        $\hat{h}^{-j}_{m,l} = \arg\min\{h \in \mathcal{H}_m : \sum_{(\boldsymbol{x},d) \in (\mathcal{P}_{m,-j})_l} c_k(h(\boldsymbol{x}), d)\}$ // Train submodels
      end for
    end for
    $L^j(\boldsymbol{\alpha}) \leftarrow 0$
    for $k = 1, 2, \ldots, K$ **do**
      for $(\boldsymbol{x}, d) \in \mathcal{S}_{k,j}$ **do**
        $\hat{d} \leftarrow \sum_{m=1}^{M} \alpha_m \cdot (\sum_{l=1}^{l_m} \hat{h}^{-j}_{m,l}(\boldsymbol{x}) \mathbb{1}_{\{\mathcal{S}_{k,-j} \in (\mathcal{P}_{m,-j})_l\}})$ // Assembly submodels for final outputs
      $L^j(\boldsymbol{\alpha}) \leftarrow L^j(\boldsymbol{\alpha}) + c_k(\hat{d}, d)$ // Compute cross-validation out-of-sample cost
      end for
    end for
  end for
end for
$\boldsymbol{\alpha}^* \leftarrow \arg\min_{\boldsymbol{\alpha} \in \mathcal{A}} \sum_{j=1}^{J} L^j(\boldsymbol{\alpha})$ // Tuning the weights of submodels

Output: $\{\hat{h}_m = (\hat{h}_{m,1}, \hat{h}_{m,2}, \ldots, \hat{h}_{m,l_m}), m \in [M]\}$, $\boldsymbol{\alpha}^*$

As demonstrated in Algorithm 1, the flexibility and generality of the FlexDA framework lie in its ability to incorporate multiple partitions of data sets, model nonlinear feature-demand relationships, and optimize general operational objective functions. Given that both demand prediction and decision making in a data-driven setting are critical for daily operations at Meituan, as well as other retailing platforms, we analyze and validate the performance of the FlexDA framework for these two core tasks. Specifically, we analytically investigate the FlexDA method with linear models and MSE cost function for prediction tasks (see Section 5). We demonstrate its advantage over the conventional data-driven approaches through theoretical analysis and extensive numerical experiments. We extend the analysis of the FlexDA method to settings with nonlinear feature-response relationships, as well as general cost functions, and establish a performance guarantee of the FlexDA method in those more complex settings (see Section 6). Finally, building on the theoretical development and empirical validation, we implemented FlexDA with Meituan to address the ordering problems in their CGB business (see Section 7).

4.2. Relation to Tree-Based and Ensemble Models

In this section, we discuss the connection and conceptual difference with related models such as regression trees and ensemble models.

4.2.1. Tree Models. The partition of data sets in FlexDA seems related to the partition of feature space in tree-based models. However, their underlying purposes are conceptually distinct. In regression trees, which are nonparametric models, the partitioning aims to divide the feature space into subregions and compute the average response within each subregion, thereby capturing nonlinear feature–response relationships. When a regression tree is applied to the entire data set across all available feature dimensions, the procedure yields a single partition of the feature space,

typically determined by heuristic splitting rules. In contrast, FlexDA considers multiple partitions, each guiding data aggregation to train a separate submodel. These submodels are then aggregated through a set of carefully tuned weights to produce the final prediction or decision.

To illustrate the advantage of employing multiple partitions in FlexDA over tree-based models, we examine a simplified data-generating process where feature information is absent and only the product index is available. Specifically, suppose there are K products, and the demand for product k follows:

$$d_k = \mu_k + \epsilon_k, \tag{6}$$

where ϵ_k is a zero-mean random noise term with variance σ_k^2. The historical data set for product k is denoted by $\mathcal{S}_k = \{\hat{d}_k^i : i \in [N]\}$. Under the MSE loss, the estimator based solely on the individual data set $\mathcal{S}_k$ is $\hat{\mu}_k = \frac{1}{N}\sum_{i=1}^N \hat{d}_k^i$. As illustrated in Figure 4, if one applies regression trees to the fully aggregated data set, the resulting tree structure splits by the product indices. If the tree depth is s, there are s splits, resulting $(s+1)$ leaves: The first s leaves correspond to products that are fully separated ("decoupled products"), and the remaining $(K-s)$ products share a pooled leaf ("pooled products"). Let the set of decoupled products be denoted by $\mathbb{S}_s$, and the set of pooled products by $\mathbb{S}_{K-s}$. Then, the regression tree model $\hat{T}_0(k)$ can be expressed as

$$\hat{T}_0(k) = \begin{cases} \hat{\mu}_k, & \text{if } k \in \mathbb{S}_s, \\ \hat{\mu}_p = \frac{1}{K-s}\sum_{k\in\mathbb{S}_{K-s}} \frac{1}{N}\sum_{i=1}^N \hat{d}_k^i, & \text{if } k \in \mathbb{S}_{K-s}. \end{cases} \tag{7}$$

We consider a two-level FlexDA estimator that combines the predictions based on individual data sets and the predictions from the regression tree model:

$$\hat{\mu}_k^{\text{FlexDA}}(\alpha) = \alpha\hat{\mu}_k + (1-\alpha)\hat{T}_0(k), \tag{8}$$

where $\alpha \in [0,1]$ denotes the weight. Obviously, when $\alpha = 1(0)$, the FlexDA estimator reduces to the individual estimator (regression tree). Under the MSE criterion, the average out-of-sample loss of the FlexDA estimator is

$$\text{MSE}(\alpha) = \frac{1}{K}\sum_{k=1}^K \mathbb{E}_{\mathcal{S}_k}[\mathbb{E}_{\epsilon_k}[(\hat{\mu}_k^{\text{FlexDA}}(\alpha) - \mu_k - \epsilon_k)^2]]. \tag{9}$$

Here, MSE(0) corresponds to out-of-sample cost of the regression tree model $T_0(k)$, and MSE(1) to that of the decoupled model $\hat{\mu}_k$.

Proposition 1. *The two-level FlexDA estimator is defined in* Equation (8). MSE(α) *denotes the out-of-sample MSE for the two-level FlexDA with a weight* α. *There exists a weight* $\alpha^* \in [0,1]$ *such that*

$$\text{MSE}(\alpha^*) \le \min\{\text{MSE}(0),\ \text{MSE}(1)\}. \tag{10}$$

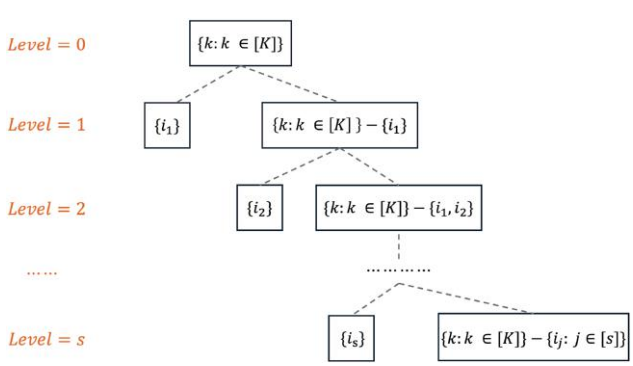
Proposition 1 demonstrates that the FlexDA estimator can outperform the individual estimator and the regression tree model trained on the fully aggregated data set by carefully combining two predictions via a carefully chosen weight.

In general, the partitioning in regression trees aims to reduce bias by grouping similar data points together, thereby capturing nonlinear feature–response relationships. In the preceding example, increasing the tree depth reduces bias but simultaneously increases model complexity, which in turn raises variance. Thus, the tradeoff between bias and variance in regression trees is governed by the choice of tree depth.

For a given tree level s such that $s < K$, there exists a subgroup of products, illustrated by the rightmost leaf node in Figure 4, that are similar in nature. However, the output of this leaf, computed as the sample mean of the demands of all products within the subgroup,[1] neglects the heterogeneity among these products. The two-level FlexDA estimator addresses this limitation by combining the unbiased sample means of individual products with the leaf-level output, which exhibits lower variance. By appropriately selecting the weighting scheme, the FlexDA estimator achieves improved overall performance through a more effective balance between bias and variance within this subgroup of products.

In sum, the multiple partitions in FlexDA serve to increase model diversity, which refers to the variation among candidate models. As shown by Krogh and Vedelsby (1994), the generalization error of a combination of submodels equals the weighted average of the individual model errors minus the weighted average ambiguity, which measures the disagreement among the models. Unlike regression trees, the partitioning in FlexDA is not intended to capture feature–response nonlinearity; rather, it is designed to intentionally generate diversity across candidate models by varying the degree of data pooling across products.

Figure 4. (Color online) Regression Tree with Product Labels as the Only Feature



4.2.2. Ensemble Models. The model in Equation (5) appears similar to ensemble learning in the machine

learning field, such as Stacking or Superlearner, which combine submodels with a collection of weights derived from cross-validation or regression (Van der Laan et al. 2007, Hastie et al. 2009). Also, the model appears similar to model averaging in statistics and econometrics, which takes the average across the outputs of multiple models (Liao et al. 2019, Li et al. 2025, Peng et al. 2025). However, FlexDA and ensemble learning differ in how to generate model diversity. The former increases model disagreement by aggregating data with purposeful data partitioning rules; the latter typically involves submodels from distinct model classes or generates different models via bootstrapped samples or random feature selection. Additionally, the existing asymptotic analysis of Stacking or Superlearner focuses on the "big-data" regime where the sample size goes to infinity (Le and Clarke 2022, Zhang and Liu 2023). Our theoretical analysis with FlexDA considers limited data samples in a large-scale system, which is the so-called small-data large-scale regime.

4.2.3. Transfer Learning. Our FlexDA framework is related to transfer learning, which improves performance for data-scarce problems by leveraging information from related domains (Wang and Mahadevan 2011, Nam and Kim 2015, Bastani 2021). Although both approaches utilize cross-product information, they differ in structural assumptions and asymptotic regimes. Transfer learning typically assumes that the source and target tasks share a latent representation, so that knowledge learned from a data-rich source can be transferred to improve a data-scarce target. In contrast, FlexDA addresses prediction and decision problems for multiple products, all of which face limited data. It does not impose a specific similarity structure across products; instead, it adapts to varying degrees of homogeneity through flexible data aggregation. Conceptually, transfer learning aims to recover shared structure across products, whereas FlexDA enhances model diversity through multiple aggregation levels and captures product homogeneity and heterogeneity via decision-aware weights assigned to submodels trained at different aggregation levels.

5. Flexible Data Aggregation (FlexDA) for Prediction

In this section, we implement the FlexDA approach for prediction tasks in a data-driven setting. In particular, we consider a two-level FlexDA estimator, which combines a shared model and decoupled models with a weight parameter. We characterize the optimal parameter and demonstrate the advantage of FlexDA estimators over conventional approaches with such an optimal weight. We further propose a data-driven estimator of the weight, which converges to the optimal one as the number of products approaches infinity. In Online Appendix B, we validate the FlexDA estimator using both synthetic and real data and verify its advantages over conventional and modern data-driven approaches.

As discussed previously, one critical component of the FlexDA approach is the weight assigned to each submodel. Under the linear structure, we characterize the theoretically optimal weights and propose consistent estimators of the weights in data-driven settings. Specifically, the random demand D_k for product k is assumed to be a function of the feature vector $\boldsymbol{x}_k$ and a random noise ϵ_k:

$$D_k = \boldsymbol{\beta}_k^\top \boldsymbol{x}_k + \epsilon_k, \tag{11}$$

where $\boldsymbol{\beta}_k$ is the true parameter vector and ϵ_k follows a Gaussian distribution with zero mean and variance σ_k^2. We further impose the following condition that the true parameter vector $\boldsymbol{\beta}_k$ is bounded.

Assumption 1. *For each product $k \in [K]$, $\boldsymbol{\beta}_k$ is randomly drawn from a distribution with a mean of $\boldsymbol{\beta}_0$, that is, $\mathbb{E}[\boldsymbol{\beta}_k] = \boldsymbol{\beta}_0$, and a bounded support, that is, $|\boldsymbol{\beta}_k|_\infty < \beta_{\max}$ where $\beta_{\max} > 0$ is a constant.*

In this section, we focus on the demand prediction problem with contextual information and thus set the cost function to the mean squared error. Given the linear structure between the feature and the demand mean, it is natural to restrict the model space to the class of linear functions, $\mathcal{H}_L = \{h(\boldsymbol{x}) = \tilde{\boldsymbol{\beta}}^\top \boldsymbol{x}, \tilde{\boldsymbol{\beta}} \in \mathbb{R}^F\}$. The firm's optimization problem is

$$\min_{\tilde{\boldsymbol{\beta}}_k, k\in[K]} \frac{1}{K} \sum_{k\in[K]} \mathbb{E}_{D_k}[(\tilde{\boldsymbol{\beta}}_k^\top \boldsymbol{x}_k - D_k)^2 | \boldsymbol{X}_k = \boldsymbol{x}_k]. \tag{12}$$

As discussed in Section 3, however, the conditional distribution of the random demand is not known. The firm instead has access to a historical data set $\mathcal{S}$ that contains the past feature vectors and the realized demands. Before we proceed to describe how to implement the FlexDA approach as well as the benchmark ones, we impose conditions on the observed data set, which are commonly assumed in the literature.

Assumption 2. (i) *For any product k, the sample feature covariance matrix $\hat{X}_k^\top \hat{X}_k$ is positive definite of which the smallest eigenvalue $\underline{\lambda} > 0$.* (ii) *There exists a constant $x_{\max}$ such that $|x_{k,j}| \le x_{\max}$ for any product $k \in [K]$ and $j \in [F]$.*

Assumption 2(i) is a common assumption in the literature to ensure that the ordinary least squares (OLS) estimator of the coefficient $\boldsymbol{\beta}_k$ is unique. Assumption 2(ii) implies that the feature value is bounded, which is not restrictive, because one can always normalize the feature. Before introducing the FlexDA approach for demand prediction with linear models, we first present the two benchmark approaches that rely solely on individual data sets and a fully aggregated data set.

5.1. Decoupled OLS Estimator

As discussed in Section 3, one can estimate demand means for each product based on respective data sets. Under the MSE cost function, this essentially gives the OLS estimators for each product:

$$\hat{\boldsymbol{\beta}}_k \in \operatorname*{argmin}_{\tilde{\boldsymbol{\beta}}_k} \frac{1}{N_k}\sum_{i=1}^{N_k}(\tilde{\boldsymbol{\beta}}_k^\top \hat{\boldsymbol{x}}_k^i - \hat{d}_k^i)^2. \tag{13}$$

Under Assumption 2, we can derive the closed-form OLS estimator as follows:

$$\hat{\boldsymbol{\beta}}_k = (\hat{X}_k^\top \hat{X}_k)^{-1}\hat{X}_k^\top \hat{D}_k, \tag{14}$$

which satisfies the following properties $\mathbb{E}[\hat{\boldsymbol{\beta}}_k] = \boldsymbol{\beta}_k$, $\mathbb{V}[\hat{\boldsymbol{\beta}}_k] = \sigma_k^2(\hat{X}_k^\top \hat{X}_k)^{-1}$. It is known that the OLS estimator $\hat{\boldsymbol{\beta}}_k$ is unbiased but subject to high variance, especially when the sample size N_k is small. For example, in the extreme case when no feature is considered, and the firm estimates the demand mean merely based on past observations, the variance of the OLS estimator is $\mathbb{V}[\hat{\boldsymbol{\beta}}_k] = \sigma_k^2/N_k$, decreasing in the sample size. Finally, we evaluate the decoupled OLS estimator $\hat{\boldsymbol{\beta}}^{\text{ols}} = \{\hat{\boldsymbol{\beta}}_k, k \in [K]\}$ by computing the average expected out-of-sample cost as follows:

$$\begin{aligned}\mathcal{L}(\hat{\boldsymbol{\beta}}^{\text{ols}}) &= \frac{1}{K}\sum_{k\in[K]} \mathbb{E}_{\mathcal{S}_k}[\mathbb{E}_{D_k}[(\hat{\boldsymbol{\beta}}_k^\top \boldsymbol{x}_k - D_k)^2 | \boldsymbol{x}_k] | \hat{X}_k] \\ &= \frac{1}{K}\sum_{k\in[K]} \boldsymbol{x}_k^\top \sigma_k^2 (\hat{X}_k^\top \hat{X}_k)^{-1}\boldsymbol{x}_k + \frac{1}{K}\sum_{k\in[K]}\sigma_k^2,\end{aligned} \tag{15}$$

where the inner expectation is taken with respect to the future random demand and the outer expectation is taken with respect to the historical data sets. As indicated by the right-hand side of Equation (15), the out-of-sample cost $\mathcal{L}(\hat{\boldsymbol{\beta}}^{\text{ols}})$ can be decomposed into two parts: The first part is solely associated with the variance of the unbiased estimators, and the second part is the irreducible cost related to the random noise. Even without model misspecification, the decoupled OLS estimator's out-of-sample performance can be inferior, especially with a limited sample size, resulting from the high variance of the estimators.

5.2. Shared OLS Estimator

One can estimate the parameter for each product by aggregating all data sets in a single one to address the issue of data scarcity (Chuang et al. 2021, Lei et al. 2025). This approach implicitly assumes that there exists a parameter vector shared by all products. Therefore, one can obtain a shared OLS estimator $\hat{\boldsymbol{\beta}}_0$ by minimizing the sum of squared errors across all data samples:

$$\hat{\boldsymbol{\beta}}_0 \in \operatorname*{argmin}_{\tilde{\boldsymbol{\beta}}} \frac{1}{\mathbb{N}}\sum_{k=1}^{K}\sum_{i=1}^{N_k}(\tilde{\boldsymbol{\beta}}^\top \hat{\boldsymbol{x}}_k^i - \hat{d}_k^i)^2. \tag{16}$$

When Assumption 2 holds, we can derive the closed-form shared OLS estimator, $\hat{\boldsymbol{\beta}}_0 = (\hat{X}^\top \hat{X})^{-1}\hat{X}^\top \hat{D}$, where $\hat{X} = [\hat{X}_1, \hat{X}_2, \ldots, \hat{X}_K]^\top \in \mathbb{R}^{\mathbb{N}\times F}$ denotes the total feature matrix and $\hat{D} = [\hat{D}_1, \hat{D}_2, \ldots, \hat{D}_K]^\top \in \mathbb{R}^{\mathbb{N}}$ denotes the total demand observations. The following lemma shows that the shared estimator is a consistent estimator of the mean of the true parameter vector.

Lemma 1. *When Assumptions 1 and 2 hold, as $K \to \infty$, we have, $\hat{\boldsymbol{\beta}}_0 \to_p \boldsymbol{\beta}_0$.*

In light of Lemma 1, in order to evaluate the shared OLS estimator, one can compute the averaged expected out-of-sample cost associated with the mean of the parameter vector, $\boldsymbol{\beta}_0$, which provides a lower bound of the out-of-sample cost associated with the shared OLS estimator:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\beta}_0) &= \frac{1}{K}\sum_{k\in[K]} \mathbb{E}_{\mathcal{S}_k}[\mathbb{E}_{D_k}[(\boldsymbol{\beta}_0^\top \boldsymbol{x}_k - D_k)^2 | \boldsymbol{x}_k] | \hat{X}_k] \\ &= \frac{1}{K}\sum_{k\in[K]} \boldsymbol{x}_k^\top (\boldsymbol{\beta}_0 - \boldsymbol{\beta}_k)(\boldsymbol{\beta}_0 - \boldsymbol{\beta}_k)^\top \boldsymbol{x}_k + \frac{1}{K}\sum_{k\in[K]}\sigma_k^2.\end{aligned}$$

Based on the above relation, the out-of-sample cost associated with the shared estimator is related to the irreducible variance of the random noise and the inherent variation across the true parameter vectors.

5.3. Two-Level FlexDA Estimator

The FlexDA method hypothetically can aggregate the data sets based on any valid partition and train the submodel accordingly. In this section, we consider a specific two-level FlexDA estimator that combines the outputs from two submodels that are trained with the no-aggregation partition and the full-aggregation partition with a weight $\alpha \in [0, 1]$:

$$\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\alpha, \hat{\boldsymbol{\beta}}_0, \hat{\boldsymbol{\beta}}_k) = \alpha\hat{\boldsymbol{\beta}}_k + (1-\alpha)\hat{\boldsymbol{\beta}}_0. \tag{17}$$

To demonstrate the effectiveness of the two-level FlexDA estimator, inspired by Lemma 1, we first consider a proxy estimator that sets the estimated shared parameter $\hat{\boldsymbol{\beta}}_0$ to true parameter mean $\boldsymbol{\beta}_0$ with the two-level FlexDA estimator, that is, $\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\alpha, \boldsymbol{\beta}_0, \hat{\boldsymbol{\beta}}_k)$. We first characterize the optimal weight associated with $\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\alpha, \boldsymbol{\beta}_0, \hat{\boldsymbol{\beta}}_k)$ and demonstrate the value of flexible data aggregation. Subsequently, we propose a data-driven method that provides a consistent estimator of the optimal weight that can enable the FlexDA estimator to achieve a smaller out-of-sample cost than the decoupled estimators or the shared one. Specifically, the average expected

out-of-sample cost with the two-level FlexDA estimator using the weight α can be computed as

$$\begin{aligned}
&\mathcal{L}^{\text{FlexDA}}(\alpha;\boldsymbol{\beta}_0)\\
&=\frac{1}{K}\sum_{k\in[K]}\mathbb{E}_{\mathcal{S}_k}[\boldsymbol{x}_k^\top(\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\alpha,\boldsymbol{\beta}_0,\hat{\boldsymbol{\beta}}_k)-\boldsymbol{\beta}_k)\\
&\quad(\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\alpha,\boldsymbol{\beta}_0,\hat{\boldsymbol{\beta}}_k)-\boldsymbol{\beta}_k)^\top\boldsymbol{x}_k\,|\,\hat{X}_k]+\frac{1}{K}\sum_{k\in[K]}\sigma_k^2\\
&=\underbrace{\alpha^2\frac{1}{K}\textstyle\sum_{k\in[K]}\boldsymbol{x}_k^\top\sigma_k^2(\hat{X}_k^\top\hat{X}_k)^{-1}\boldsymbol{x}_k}_{\text{Variance}}\\
&\quad+\underbrace{(1-\alpha)^2\frac{1}{K}\textstyle\sum_{k\in[K]}\boldsymbol{x}_k^\top(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)^\top\boldsymbol{x}_k}_{\text{Bias}}\\
&\quad+\frac{1}{K}\sum_{k\in[K]}\sigma_k^2.
\end{aligned}$$

Similar to the out-of-sample cost associated with the decoupled estimators, $\mathcal{L}(\hat{\boldsymbol{\beta}}^{\text{ols}})$, the first cost component of $\mathcal{L}^{\text{FlexDA}}(\alpha;\boldsymbol{\beta}_0)$ is associated with the variance of OLS estimators, which is reduced with the weight $\alpha\in[0,1]$. On the other hand, the second cost component arises due to the bias introduced by including the shared parameter $\boldsymbol{\beta}_0$. The choice of α critically affects the interplay between the reduced variance and the increased bias. The following proposition formally characterizes the optimal choice of the weight given the shared parameter and showcases the advantage of the FlexDA approach over the conventional approaches.

Proposition 2. *When Assumption 2 holds, given the anchor parameter $\boldsymbol{\beta}_0$, if the weight α satisfies the following:*

$$\alpha^*=\frac{\sum_{k\in[K]}\boldsymbol{x}_k^\top(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)^\top\boldsymbol{x}_k}{\sum_{k\in[K]}\boldsymbol{x}_k^\top[\mathbb{E}_{\mathcal{S}_k}[(\hat{\boldsymbol{\beta}}_k-\boldsymbol{\beta}_k)(\hat{\boldsymbol{\beta}}_k-\boldsymbol{\beta}_k)^\top]+(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)^\top]\boldsymbol{x}_k},$$

we have, $\alpha^*\in(0,1)$ and the averaged out-of-sample cost $\mathcal{L}^{\text{FlexDA}}(\alpha^*;\boldsymbol{\beta}_0)$ is minimized. Furthermore, we have

$$\mathcal{L}(\hat{\boldsymbol{\beta}}^{\text{ols}})-\mathcal{L}^{\text{FlexDA}}(\alpha^*;\boldsymbol{\beta}_0)=\frac{1-\alpha^*}{K}\sum_{k\in[K]}\boldsymbol{x}_k^\top\sigma_k^2(\hat{X}_k^\top\hat{X}_k)^{-1}\boldsymbol{x}_k>0,$$

$$\mathcal{L}(\boldsymbol{\beta}_0)-\mathcal{L}^{\text{FlexDA}}(\alpha^*;\boldsymbol{\beta}_0)=\frac{\alpha^*}{K}\sum_{k\in[K]}\boldsymbol{x}_k^\top(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)^\top\boldsymbol{x}_k>0.$$

Proposition 2 highlights that, with an appropriate choice of weight α^*, the FlexDA estimator that combines the unbiased decoupled estimators, and the shared one outperforms the decoupled estimators or the shared estimator alone. Note that the expression of α^* in Proposition 2 implicitly reveals the impact of sample size N_k. It is known that, as the sample size N_k increases, the variance of decoupled OLS estimators, $\mathbb{E}_{\mathcal{S}_k}[(\hat{\boldsymbol{\beta}}_k-\boldsymbol{\beta}_k)(\hat{\boldsymbol{\beta}}_k-\boldsymbol{\beta}_k)^\top]$, decreases. As a result, the larger the sample size, the larger α^* is, and thus, the advantage of the FlexDA estimator over the decoupled approach diminishes, because the decoupled estimator becomes more accurate with more data. Conversely, a larger sample size increases the benefit of the two-level FlexDA estimator over the shared OLS estimator, because the decoupled OLS estimator becomes more effective and receives a larger weight.

However, the calculation of the optimal weight α^* requires the true parameters, $\boldsymbol{\beta}_1,\ldots,\boldsymbol{\beta}_K$, as well as the true parameter mean $\boldsymbol{\beta}_0$, which are not known. Thus, we propose a consistent estimator of α^* to derive the weight for the FlexDA estimator in a data-driven setting.

Proposition 3. *Given the entire data set $\mathcal{S}=\{\mathcal{S}_1,\ldots,\mathcal{S}_k\}$, the weight is computed as follows:*

$$\hat{\alpha}=\frac{\frac{1}{K}\sum_{k\in[K]}\boldsymbol{x}_k^\top(\hat{\boldsymbol{\beta}}_k-\hat{\boldsymbol{\beta}}_0)(\hat{\boldsymbol{\beta}}_k-\hat{\boldsymbol{\beta}}_0)^\top\boldsymbol{x}_k-\frac{1}{K}\sum_{k\in[K]}s_k^2\boldsymbol{x}_k^\top(\hat{X}_k^\top\hat{X}_k)^{-1}\boldsymbol{x}_k}{\frac{1}{K}\sum_{k\in[K]}\boldsymbol{x}_k^\top(\hat{\boldsymbol{\beta}}_k-\hat{\boldsymbol{\beta}}_0)(\hat{\boldsymbol{\beta}}_k-\hat{\boldsymbol{\beta}}_0)^\top\boldsymbol{x}_k},$$

where $s_k^2=\frac{(\hat{D}_k-\hat{X}_k\hat{\boldsymbol{\beta}}_k)^\top(\hat{D}_k-\hat{X}_k\hat{\boldsymbol{\beta}}_k)}{N_k-F}$. When Assumption 2 holds, as $K\to\infty$, we have, $\hat{\alpha}\to_p\alpha^*$.

Proposition 3 implies that, as the number of products goes to infinity, the data-driven weight $\hat{\alpha}$ converges to the optimal weight, that can be obtained with the true distribution information. This provides a performance guarantee on the data-driven estimator that we proposed. Furthermore, a direct observation generated from Proposition 3 is that $\mathbb{E}[\bar{\boldsymbol{\beta}}_k^{\text{FlexDA}}]=\alpha^*\boldsymbol{\beta}_k+(1-\alpha^*)\boldsymbol{\beta}_0$, and $\mathbb{V}[\bar{\boldsymbol{\beta}}_k^{\text{FlexDA}}]=(\alpha^*)^2\mathbb{V}[\hat{\boldsymbol{\beta}}_k]$, where $\bar{\boldsymbol{\beta}}_k^{\text{FlexDA}}=\lim_{K\to\infty}\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\hat{\alpha},\hat{\boldsymbol{\beta}}_0,\hat{\boldsymbol{\beta}}_k)$. Compared with the decoupled OLS estimator defined in Equation (14), the two-level FlexDA estimator encounters a lower variance as $\alpha^*\in(0,1)$ but bears an increased bias $(1-\alpha^*)(\boldsymbol{\beta}_0-\boldsymbol{\beta}_k)$, depending on how close the true parameters $\boldsymbol{\beta}_k$ are to the mean $\boldsymbol{\beta}_0$. This illustrates how the FlexDA approach can balance the bias–variance tradeoff via data aggregation and a tunable weight, which can potentially generate superior performance, especially in the small-data large-scale regime. Next, we demonstrate the effectiveness of the data-driven parameters $\hat{\alpha}$ by comparing the associated out-of-sample cost with that of the optimal weight and true parameter mean.

Theorem 1. *Given the data set $\mathcal{S}$, when Assumption 2 holds, as $K\to\infty$, we have*

$$\begin{aligned}
&\frac{1}{K}\sum_{k\in[K]}(\boldsymbol{x}_k^\top(\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\hat{\alpha},\hat{\boldsymbol{\beta}}_0,\hat{\boldsymbol{\beta}}_k)-\boldsymbol{\beta}_k)\\
&\quad(\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\hat{\alpha},\hat{\boldsymbol{\beta}}_0,\hat{\boldsymbol{\beta}}_k)-\boldsymbol{\beta}_k)^\top\boldsymbol{x}_k)\\
&-\frac{1}{K}\sum_{k\in[K]}(\boldsymbol{x}_k^\top(\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\alpha^*,\boldsymbol{\beta}_0,\hat{\boldsymbol{\beta}}_k)-\boldsymbol{\beta}_k)\\
&\quad(\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\alpha^*,\boldsymbol{\beta}_0,\hat{\boldsymbol{\beta}}_k)-\boldsymbol{\beta}_k)^\top\boldsymbol{x}_k)\to_p 0.
\end{aligned}$$

Theorem 1 shows that, as the number of products K increases, the out-of-sample cost associated with the

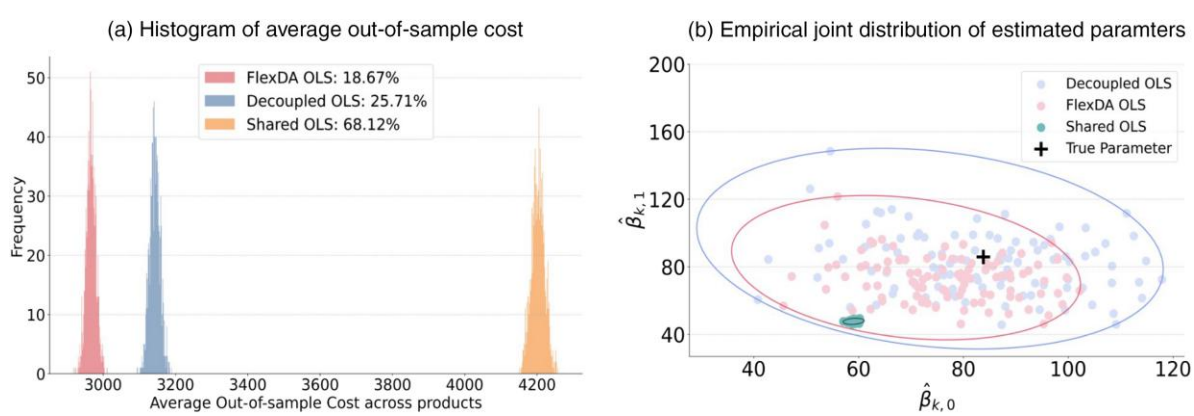
FlexDA estimator $\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\hat{\alpha}, \hat{\boldsymbol{\beta}}_0, \hat{\boldsymbol{\beta}}_k)$ converges to the out-of-sample cost associated with $\hat{\boldsymbol{\beta}}_k^{\text{FlexDA}}(\alpha^*, \boldsymbol{\beta}_0, \hat{\boldsymbol{\beta}}_k)$ that utilizes the true means to tune the weight. This indicates that in the small-data large-scale regime, the cost incurred by the estimated weight and parameters approaches that of the optimal ones. Thus, combining the results of Proposition 2 and Theorem 1, we can conclude that, when the number of products is large, using the estimated weight $\hat{\alpha}$ and anchor parameter $\hat{\boldsymbol{\beta}}_0$, the FlexDA approach outperforms the conventional approaches such as decoupled OLS estimators as well as the shared estimator, especially with limited sample sizes.

We further demonstrate the effectiveness of the proposed FlexDA method (denoted as "FlexDA OLS" hereafter) compared with two benchmark approaches via numerical experiments with synthetic data and unfold the intuition of why the FlexDA method can improve the bias–variance tradeoff. The first benchmark trains a linear model for each product using respective data sets, representing the no-aggregation approach (denoted as "decoupled OLS" hereafter). The second benchmark estimates the shared parameter with the data aggregated from all products, representing the full-aggregation approach (denoted as "shared OLS" hereafter). In order to highlight the properties of the proposed FlexDA method, we focus on comparing the FlexDA method with these two benchmarks with a linear feature–demand relationship. For each product k, the demand mean is a linear function of a one-dimensional feature. Specifically, $D_k = \beta_{k,0} + \beta_{k,1}x_k + \epsilon_k$, where the random noise $\epsilon_k \sim \mathcal{N}(0, 50^2)$. The parameters $\beta_{k,0}$ and $\beta_{k,1}$ are randomly drawn from a uniform distribution spanning [0, 100]. The feature of product k, X_k, is randomly sampled from the standard normal distribution, $\mathcal{N}(0,1)$. The number of products K is set to 1,000, and the sample size for each product N_k is set to 10. We report the average out-of-sample costs across products in the histogram, based on 100 simulated instances, as demonstrated in Figure 5(a).

The proposed FlexDA method consistently achieves the lowest out-of-sample costs and outperforms the benchmarks in all simulated scenarios. Surprisingly, even the worst-case out-of-sample costs obtained by the FlexDA method is lower than the best-case ones achieved by the two benchmarks. This demonstrates the superiority and robustness of the proposed FlexDA method. We further compute the optimality gaps of each method by comparing their out-of-sample costs with the optimal ones. The FlexDA method achieves an average optimality gap of 18.67%. In comparison, the decoupled OLS and shared OLS approaches result in higher expected optimality gaps of 25.71% and 68.12%, respectively. The substantial optimality gaps resulting from the shared OLS approach underscore the importance of capturing the heterogeneity among products. The advantage of the FlexDA method over the decoupled OLS highlights the value of modeling homogeneity among products via data aggregation.

Figure 5(b) reports the distribution of estimated parameters $\hat{\beta}_{k,0}$ and $\hat{\beta}_{k,1}$, with smallest ellipses that cover all pairs of estimates for each method. Decoupled OLS results in average estimates that are closer to the true ones compared with the other two methods, but its possible estimates spread over the broadest ranges. This suggests that the decoupled approach has the lowest bias but suffers from the largest variance for parameter estimation. On the contrary, the shared OLS has the narrowest range of possible estimates, but the average estimates significantly deviate from the true ones. This confirms the intuition that aggregating data helps reduce the variance at the cost of introducing bias. The proposed FlexDA method, which integrates the complementary advantages of two benchmarks, results in a

Figure 5. (Color online) Histogram of Average Out-of-Sample Costs with 100 Randomly Generated Instances



Notes. (a) The x axis denotes the average out-of-sample cost across products, and the y axis denotes the frequency. The percentages in the legend denote the average optimality gap ratio for each method, computed across 100 instances. (b) Each dot of a particular color corresponds to the coefficient estimates from one instance under the respective method.

moderate range of possible estimates and limited deviation from the true parameters. By effectively reducing variance while bearing an acceptable increased bias, the proposed FlexDA method should provide accurate and robust predictions and solutions with limited data in practice. We further conduct extensive numerical experiments on both synthetic and real data to demonstrate the effectiveness of FlexDA for prediction. Detailed results are provided in Online Appendix B.1 and Online Appendix B.2.

6. Flexible Data Aggregation (FlexDA) for Decision Making

The theoretical results presented in Section 5 establish a performance guarantee for the FlexDA approach with linear models for prediction tasks. In this section, we extend our analysis to a more general setting that encompasses broader model classes and general cost functions. Specifically, we assume that the random demand possesses a nonlinear relationship with the feature vector. We first theoretically investigate the performance of the FlexDA approach under general settings by characterizing the upper bound of the suboptimality gap, which converges to zero when the number of products is large. With the established performance guarantee, we implement the proposed FlexDA approach to manage a newsvendor system with both synthetic and real data (see Online Appendix B). Extensive numerical studies demonstrate the effectiveness of the proposed approach and highlight its advantage over the existing approaches.

In particular, we extend the analysis to the case with nonlinear feature–demand relationships. In particular, the data-generating process for the random demand of product k is given as

$$D_k = g_k(\boldsymbol{x}_k) + \epsilon_k, \tag{18}$$

where $g_k \in \mathcal{G}$ is a general function mapping the feature vector to the true demand mean and ϵ_k denotes the random noise. Furthermore, because we implement the FlexDA method to derive data-driven decisions, the objective function c_k is not necessary in the form of MSE but can be in a general form, such as the newsvendor cost function, depending on the operational characteristics of application settings.

We construct a two-level FlexDA model, consisting of a shared model trained with a fully aggregated data set and decoupled models trained with individual data sets, to prescribe the data-driven decisions for all products. In particular, the decoupled model for product k, denoted by $\hat{h}_k$, can be learned by minimizing the objective on the individual data sets:

$$\hat{h}_k(\cdot) \in \operatorname*{argmin}_{h\in\mathcal{H}} \frac{1}{N_k}\sum_{i=1}^{N_k} c_k(h(\hat{\boldsymbol{x}}_k^i), \hat{d}_k^i). \tag{19}$$

The shared model that is trained with the fully aggregated data set, denoted by $\hat{h}_0$, is given by

$$\hat{h}_0(\cdot) \in \operatorname*{argmin}_{h\in\mathcal{H}} \frac{1}{\sum_{k\in[K]} N_k}\sum_{k=1}^{K}\sum_{i=1}^{N_k} c_k(h(\hat{\boldsymbol{x}}_k^i), \hat{d}_k^i). \tag{20}$$

Provided the shared and decoupled models, we can construct a two-level FlexDA model with a weight parameter $\alpha \in [0,1]$, which is expressed as

$$\hat{h}_k^{\text{FlexDA}}(\boldsymbol{x};\alpha,\hat{h}_k,\hat{h}_0) = \alpha\hat{h}_k(\boldsymbol{x}) + (1-\alpha)\hat{h}_0(\boldsymbol{x}). \tag{21}$$

For a given weight parameter α, the total out-of-sample cost of the proposed two-level FlexDA model is computed as follows:

$$Z(\alpha) = \frac{1}{K}\sum_{k=1}^{K} Z_k(\alpha), \text{ where}$$
$$Z_k(\alpha) = \mathbb{E}[c_k(\hat{h}_k^{\text{FlexDA}}(\boldsymbol{x}_k;\alpha,\hat{h}_k,\hat{h}_0), D_k)].$$

Because of the nonlinearity in feature-demand relationships and general cost functions, the optimal choice of weight parameter cannot be derived in a closed form. As described in Algorithm 1, we adopt the leave-one-out (LOO) validation to obtain the data-driven weight parameter $\hat{\alpha}$. To validate such a choice of weight parameter, we define an oracle, which can tune the parameter with the true distribution information, as a benchmark model. In particular, the oracle selects the optimal weight parameter, denoted by α^{OR}, based on the out-of-sample cost of the two-level FlexDA method:

$$\alpha^{\text{OR}} \in \operatorname*{argmin}_{\alpha\in[0,1]} Z(\alpha). \tag{22}$$

Obviously, the oracle's optimal weight α^{OR} results in the lowest possible out-of-sample cost one can achieve within the classes of two-level FlexDA models. Therefore, we can measure the performance of the two-level FlexDA models with data-driven weights by computing the following suboptimality gap:

$$\text{SubOpt}_K(\alpha) = \frac{1}{K}\sum_{k\in[K]} Z_k(\alpha) - \frac{1}{K}\sum_{k\in[K]} Z_k(\alpha^{\text{OR}}). \tag{23}$$

Before we present the formal result that characterizes the suboptimality of the two-level FlexDA method, we impose certain conditions for technical tractability, which are relevant in practice and commonly used in the literature.

Assumption 3. *We make the following assumptions:*

(i) ***(Random amount of data)*** *There exists a positive number N such that $\hat{N}_k \sim \text{Poisson}(N)$ for $k \in [K]$.*

(ii) ***(Discrete compact support)*** *Let ϵ_k follow a discrete bounded distribution with $\eta_{k,\epsilon}$ possible values. Similarly, for each component $x_{k,i}$ of $\boldsymbol{x}_k$, we assume it follows a discrete bounded distribution with $\eta_{k,i}$ possible values. Consequently,*

there exists a value η *such that* $\eta = \max_{k\in[K]}(\eta_{k,1}\cdot\eta_{k,2}\cdots\eta_{k,F}\cdot\eta_{k,\epsilon})$.

(iii) ***(Bounded and convex decision set)*** *For any product k, given feature vector* $\boldsymbol{x}_k$*, the decision set* $\boldsymbol{Q}(\boldsymbol{x}_k) = \{g(\boldsymbol{x}_k) : g \in \mathcal{H}\}$ *is convex, and there exists a constant* Π_1*, such as* $|\sup_{q\in\boldsymbol{Q}(\boldsymbol{x}_k)} q| \le \Pi_1$.

(iv) ***(Lipschitz, convex, and bounded cost)*** *For any product k, given any demand* d_k *and feature vector* $\boldsymbol{x}_k$*, there exists a constant L such that* $c_k(q_k, d_k)$ *are* $L-$ *Lipschitz and convex over* $q_k \in \boldsymbol{Q}(\boldsymbol{x}_k)$*. Furthermore, there exists a constant C such that* $\sup_{q_k\in\boldsymbol{Q}(\boldsymbol{x}_k)} c_k(q_k, d_k) \le C$ *for all k.*

Assumption 3(i) assumes that the total number of observations of each problem is random. This phenomenon commonly arises in Meituan's daily operations. Because of frequent updates and rotations of the assortment, most products are not available for sale every day. As a result, when aggregating sales data over two weeks, the number of observations per product is uncertain. Such an assumption is also common in the literature (Chou et al. 2025). Assumption 3(ii) assumes that the random feature vectors and noises are supported on discrete compact sets. In practice, decision makers often discretize the feature values to speed up the training process or enhance the interpretability of the trained model. Assumption 3(iii) assumes that given a specific feature vector $\boldsymbol{x}_k$, the decision set generated by the model class $\mathcal{H}$ is bounded and convex. Assumption 3(iv) assumes the cost function $c_k(q_k, d_k)$ is L-Lipschitz and convex over the decision set. These boundedness and continuity conditions are common in the literature (Bertsimas and Kallus 2020, Duan and Wang 2023).

Theorem 2. *When Assumption 3 holds, let* $\hat{\alpha}$ *denote the data-driven weight obtained with Algorithm 1 for a two-level FlexDA model, then for any* $0 < \delta \le 1$*, with probability at least* $1-\delta$*, we have*

$$\mathrm{SubOpt}_K(\hat{\alpha}) \le 432\sqrt{2}e^{2.5}C\left(\frac{\sqrt{\pi}}{2} + \sqrt{\log\left(1+\frac{4L\Pi_1}{C}\right)}\right)\frac{\log^{3/2}(2/\delta)\log(K)}{K}.$$

We characterize the suboptimality gap of the two-level FlexDA models with a data-driven weight $\hat{\alpha}$ with finite observations for each product. In particular, the suboptimality gap decays near-linearly (up to a logarithmic factor) in the number of products K with high probability. By the Borel-Cantelli lemma, Theorem 2 implies that the suboptimality gap converges to zero almost surely as K goes to infinity. This suggests the FlexDA model obtained by Algorithm 1 asymptotically achieves the best out-of-sample performance among all possible two-level FlexDA models. Therefore, these results theoretically validate the effectiveness of the proposed approach in small-data large-scale regimes and support its applicability to solving the replenishment problem faced by Meituan. We further conduct extensive numerical experiments with both synthetic and real data to demonstrate the effectiveness of FlexDA for decision making compared with state-of-the-art data-driven approaches. Detailed results are provided in Online Appendix B.3 and Online Appendix B.4.

7. Implementation of FlexDA at Meituan

In this section, we provide details of an internal study with Meituan on solving ordering problems for their CGB business. In Section 7.1, we show how to adapt the FlexDA method to Meituan's practices based on its operational characteristics. In Section 7.2, we present the results of applying the FlexDA method at Meituan.

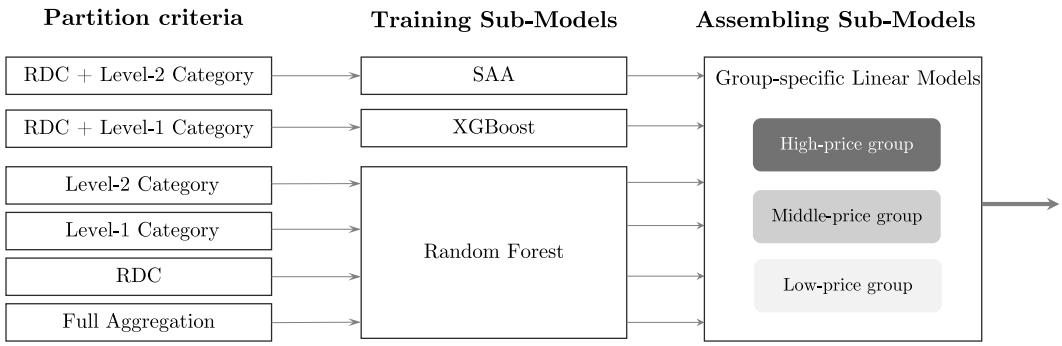
7.1. Building FlexDA Models at Meituan

The complete modeling framework for the implemented FlexDA method is illustrated in Figure 6. The implementation of the FlexDA method for Meituan's ordering problem consists of two stages. The first stage involves training different submodels using data sets aggregated at various levels, and the second stage involves assembling the outcomes of these submodels to generate the final order quantity.

One of the key features of the FlexDA method is its modeling flexibility. It can incorporate multiple data partitions and select model classes of varying complexity for each partition. To implement this method at Meituan and enhance its performance, we have made the following changes to the basic FlexDA model: (1) prescribing various levels of data aggregation and model types for each level, (2) incorporating different training objectives within a single model, and (3) assigning group-specific weights to different sets of submodels. We present these modifications in detail as follows.

7.1.1. Specifications of Data Aggregation Levels and Model Classes. In the previous analysis, we demonstrated the effectiveness of the FlexDA method with two extreme levels of data aggregation: *no aggregation*, which uses individual product data only, and *full aggregation*, which uses all the data from the entire product set. The careful combination of two-level data aggregations achieves an improved bias–variance tradeoff. Inspired by our theoretical development, firms can utilize more product information to determine different levels of data aggregation in practice. That is, if a firm can group products with similar demand patterns together, then training submodels based on such aggregated data sets can be highly effective. In addition, the flexibility of FlexDA lies not only in its ability to incorporate multiple levels of data partitions but also in its capacity to select different

Figure 6. Implemented FlexDA Model for Meituan



model classes for each partition. However, because of computational constraints, it is impractical to include all possible model classes at every partition level. Instead, we select the most appropriate model class for each partition, treating this as a supervised learning task. In particular, at Meituan, we select a model type for each partition from three major model classes: linear models, tree-based models, and deep neural networks.

As discussed in Section 4 and demonstrated in Figure 3, products at each RDC are grouped into 4 level 1 categories and 14 level 2 categories. Thus, at Meituan, we implement different levels of data aggregation by partitioning products according to the location (i.e., RDC) and category information of SKUs. The specific criteria are summarized in Figure 6 and described in detail as follows.[2]

Partition 1: RDC + level 2 category. We pool together data associated with products located in the same RDC and belonging to the same level 2 category and train a shared submodel for pooled products. Under this partition, although the data volume for some pooled data sets is relatively small, the number of features remains large. For this reason, we adopt the SAA approach to exploit the aggregated observations, disregarding the feature information.

Partition 2: RDC + level 1 category. We pool together data on products located in the same RDC and belonging to the same level 1 category, training a shared submodel for the pooled products. For this partition, we selected XGBoost as the submodels to reduce bias in demand predictions because the sample size is relatively large under level 1 category aggregation.

Partition 3: Level 2 category. We pool together data on products belonging to the same level 2 category across all RDCs, training a shared submodel for the pooled products. For this partition, each pooled subdataset contains a relatively large amount of data. Therefore, we train a random forest with the pooled data to build submodels.

Partition 4: Level 1 category. We pool together data on products belonging to the same level 1 category across all RDCs, training a shared submodel for the pooled products. Similar to Partition 3, we employ a random forest for each subdataset, which helps reduce variance.

Partition 5: RDC. We pool together data on products located in the same RDC and train a shared submodel for these products. Similar to Partitions 3 and 4, we employ a random forest for training.

Partition 6: Full aggregation. We pool together data from all products across all RDCs and train a shared submodel for the aggregated data set. Similar to Partitions 3, 4, and 5, we employ a random forest for training.

7.1.2. Incorporating Different Training Objectives Within a Single Model. Meituan adopts two performance metrics to evaluate the effectiveness of its ordering algorithms. Let p_k and q_k denote the price and order quantity for product k, respectively. The first metric is LSR, which is the ratio of the total loss of revenue due to lost sales to the maximum potential revenue. The second metric is IR, which is the ratio of leftover inventory to total demand. Two key metrics are defined as follows:

$$\mathrm{LSR} = \frac{\sum_{k=1}^{K} p_k (d_k - q_k)^+}{\sum_{k=1}^{K} p_k d_k}, \quad \mathrm{IR} = \frac{\sum_{k=1}^{K} (q_k - d_k)^+}{\sum_{k=1}^{K} d_k}. \tag{24}$$

Note that the metrics defined in Equation (24) are related to the underage and overage costs in the classical newsvendor problem. However, Meituan focuses on relative ratios rather than absolute costs. As a result, the random product demands appear in both the numerators and denominators of the LSR and IR metrics. To address this, the supply chain algorithm team at Meituan employs a positive parameter λ to combine the two metrics into a single decision objective function, $\lambda \cdot \mathrm{IR} + \mathrm{LSR}$, where a larger (smaller) value of λ places greater emphasis on the IR (LSR) metric.

As described in Figure 6, the implementation of the FlexDA method for Meituan's ordering problem consists of two stages. The first stage involves training

different submodels and the second stage involves assembling the outcomes of these submodels. Although it is critical that the training objective in the second stage aligns with the decision objective function, the choice of training objectives for the submodels in the first stage can be flexible. Moreover, Meituan may need to experiment with different values of λ to accommodate various operational settings and product categories. If the training objective were strictly tied to the decision objective function, which itself depends on λ, then testing multiple λ values would substantially increase training time. This issue is further exacerbated by the already time-consuming process of training a large number of submodels in the first stage. Therefore, when implementing the FlexDA method at Meituan, we selected MSE as the training objective in the first stage, which is now independent of the choice of λ.

In addition to the predicted demand means, we can also obtain a specific quantile of residual errors for each product under each submodel, which can further contribute to the final outputs. Specifically, for any partition m, we have subdatasets denoted by $(\mathcal{P}_m)_1, \ldots, (\mathcal{P}_m)_{l_m}$ and estimated submodels $\hat{f}_{m,1}(\cdot), \ldots, \hat{f}_{m,l_m}(\cdot)$; then, the set of residuals for any subdataset $(\mathcal{P}_m)_l$ can be represented as

$$\mathcal{R}_l^m = \{\hat{d}_{k,z} - \hat{f}_{m,l}(\hat{\boldsymbol{x}}_{k,z}) : \mathcal{S}_k \in (\mathcal{P}_m)_l, z \in [N_k]\}. \tag{25}$$

The purpose of $\mathcal{R}_l^m$ is to calculate the corresponding quantile values based on λ and the prices of different products. Using a slight abuse of notation, we denote $\mathcal{R}_l^m(\omega)$ as the ω-quantile value of the set $\mathcal{R}_l^m$ for any parameter $\omega \in (0,1)$. Thus, for any product k with feature vector $\boldsymbol{x}_k$, the output based on partition $\mathcal{P}_m$ consists of two values. The first is the predicted demand mean, that is, $\sum_{l=1}^{l_m} \mathbb{1}_{\mathcal{S}_k \in (\mathcal{P}_m)_l} \cdot \hat{f}_l^m(\boldsymbol{x}_k)$, and the second is the quantile of the corresponding residual set, that is, $\sum_{l=1}^{l_m} \mathbb{1}_{\mathcal{S}_k \in (\mathcal{P}_m)_l} \cdot \mathcal{R}_l^m\left(\frac{p_k}{\lambda + p_k}\right)$, where the intuition of setting $\omega = \frac{p_k}{\lambda + p_k}$ comes from the optimal critical quantile value in the newsvendor problem.[3]

In the second stage, as described in Figure 6, given the predicted demand means and the residual quantiles from the submodels, we then set $\lambda \cdot \mathrm{IR} + \mathrm{LSR}$ as the loss function and trained linear meta-models to obtain the submodels' weights.

7.1.3. Group-Specific Weights. In Section 4, the weight of each submodel associated with a specific level of data aggregation is assumed to be the same across all products. To introduce greater flexibility, we can partition the products into distinct groups such that model weights are shared within each group but allowed to differ across groups. This extension is particularly useful for Meituan. In particular, fresh food and standard products inherently have different holding costs; equivalently, the parameter λ should vary significantly between these two product types. Therefore, we conduct the second-stage training separately for fresh food and standard products, resulting in two distinct sets of weights for the two product types. Secondly, as shown in Besbes and Mouchtaki (2023), the sample size required for estimating different quantiles can vary significantly. The closer a quantile is to zero or one, the larger the sample size required to achieve the same level of regret, compared with a quantile near 0.5. In other words, estimating a critical quantile closer to 0.5 requires substantially fewer samples. Consequently, submodels associated with such products can perform well even with a low degree of data aggregation (e.g., no aggregation). Therefore, because the critical ratios depend on product prices, the model weights for products with medium prices should differ from those for products with relatively high or low prices. This observation suggests that a more effective implementation of FlexDA should assign group-specific weights to products with similar prices. Specifically, let $p_\tau^{1/3}$ and $p_\tau^{2/3}$ denote the one-third and two-thirds price quantiles, based on the prices of all fresh food products if $\tau = 0$ or all standard products if $\tau = 1$. For each product type τ, we then divide the products into three subgroups corresponding to the sets of products associated with low prices, middle prices, and high prices, respectively. In the linear meta-model, we assume the order quantity for product k of type τ follows the form:

$$\begin{aligned} q_k = {} & \mathbb{1}_{\{p_k \le p_\tau^{1/3}\}} \cdot \sum_{m=1}^{M} \sum_{l=1}^{l_m} \mathbb{1}_{\{\mathcal{S}_k \in (\mathcal{P}_m)_l\}} \\ & \cdot \left(\alpha_{m,1}^{\text{low}} \cdot \hat{f}_l^m(\boldsymbol{x}_k) + \alpha_{m,2}^{\text{low}} \cdot \mathcal{R}_l^m\left(\frac{p_k}{\lambda + p_k}\right)\right) \\ & + \mathbb{1}_{\{p_\tau^{1/3} < p_k \le p_\tau^{2/3}\}} \cdot \sum_{m=1}^{M} \sum_{l=1}^{l_m} \mathbb{1}_{\{\mathcal{S}_k \in (\mathcal{P}_m)_l\}} \\ & \cdot \left(\alpha_{m,1}^{\text{middle}} \cdot \hat{f}_l^m(\boldsymbol{x}_k) + \alpha_{m,2}^{\text{middle}} \cdot \mathcal{R}_l^m\left(\frac{p_k}{\lambda + p_k}\right)\right) \\ & + \mathbb{1}_{\{p_\tau^{2/3} < p_k\}} \cdot \sum_{m=1}^{M} \sum_{l=1}^{l_m} \mathbb{1}_{\{\mathcal{S}_k \in (\mathcal{P}_m)_l\}} \\ & \cdot \left(\alpha_{m,1}^{\text{high}} \cdot \hat{f}_l^m(\boldsymbol{x}_k) + \alpha_{m,2}^{\text{high}} \cdot \mathcal{R}_l^m\left(\frac{p_k}{\lambda + p_k}\right)\right). \end{aligned} \tag{26}$$

7.2. Implementation and Evaluation of FlexDA at Meituan

To evaluate the performance of the FlexDA method for decision making at Meituan, we generate order quantities and validate the decisions using real data from the company. In particular, we use 23 days of sales data,

from August 19, 2023, to September 10, 2023, from a province in China. This data set includes approximately 250,000 records, each with 326 features, collected at five RDCs in this province. When the same product is stored across different RDCs, it is treated as a distinct SKU, resulting in an average daily SKU count of around eleven thousand. Additionally, products are grouped into 4 level 1 categories and are further divided into 14 level 2 categories. This collected data set is used to evaluate and compare the performance of the proposed method and the existing one implemented at Meituan. A summary of the data set is presented in Table 1.

We employ the *rolling validation* to evaluate the performance of implemented approaches. However, strictly following the leave-one-out cross-validation in Algorithm 1 would result in prohibitively high computational costs. Therefore, we adopt the hold-out validation to expedite the implementation of Algorithm 1. Specifically, suppose the time point for which we need to make decisions is denoted as t. We first use the data points from $t-16$ to $t-3$ to train the submodels for each partition. Then, we use the data points from $t-1$ and $t-2$ to tune the weights of the submodels associated with each partition, as illustrated in Equation (26). Consequently, the test set is limited to the data from September 4, 2023, to September 10, 2023.

At Meituan, the supply chain algorithm team follows a two-step process for making ordering decisions for each product. In the first step, data of all products are aggregated, and a shared model is trained to predict multiple quantiles of the product demand distribution. Initially, XGBoost was used for predicting quantiles, but the platform has now transitioned to a transformer-based model. In the second step, the platform determines the order quantities by stochastic optimization for each product based on different objectives related to IR and LSR. The hyperparameter was $\lambda = 0.18$ for fresh products and $\lambda = 0.10$ for standard products, as specified by Meituan's supply chain algorithm team.

We compare the performance of the FlexDA model and the existing algorithm at Meituan for both fresh products and standard products, with a primary focus on decision making related to new products that lack historical data. Because of the confidentiality requirements of Meituan, we cannot disclose the absolute values of the IR and LSR metrics. Instead, we present the relative performance of the algorithms. The average relative metrics over one week are reported in Table 2. For fresh products, compared with the existing algorithm implemented by Meituan, the FlexDA model reduced IR by 42% and achieved a reduction of 14% with respect to LSR. For standard products, the FlexDA model achieved a reduction of 15% for IR and reduced LSR by 13%. This indicates that the FlexDA model significantly improves both IR and LSR metrics for both fresh and standard products. Although Meituan employs a complex transform-based model, which is trained on a fully aggregated data set and shared across all products, FlexDA delicately combines models with varying data aggregation levels and complexities to improve the bias–variance tradeoff, thereby achieving superior out-of-sample performance.

Table 1. Descriptive Statistics of the Data Set at Meituan

Variable	Value
Time span	August 19, 2023–September 10, 2023
Number of RDCs	5
Average number of SKUs sold per day	≈11,000
Number of level 1 categories	4
Number of level 2 categories	14
Number of features	326
Total data size	≈0.25 million

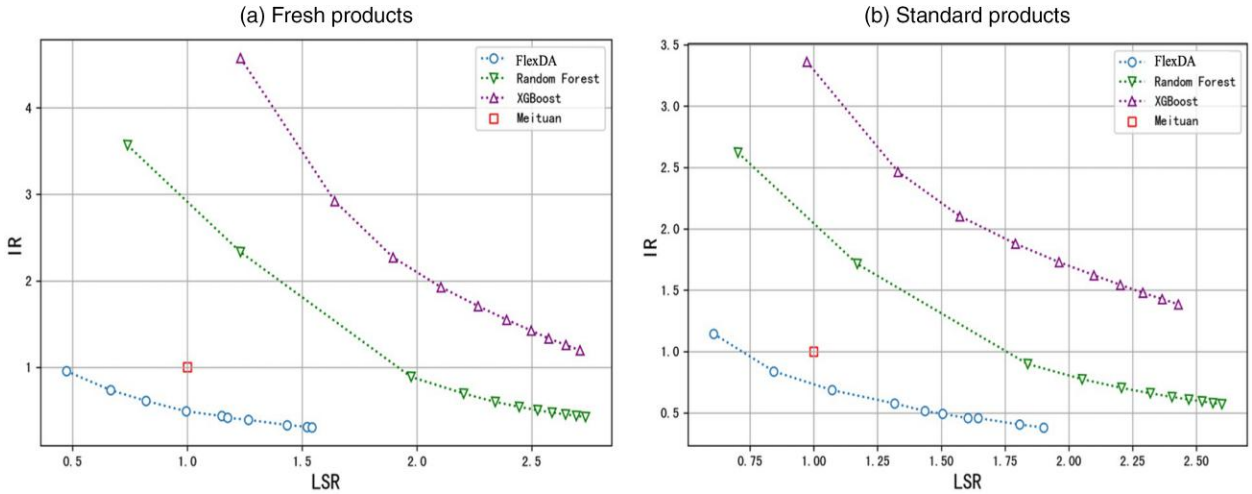
Table 2. Relative Average Performance Metrics

Product type	λ	$\frac{\mathrm{IR}_{\mathrm{FlexDA}}}{\mathrm{IR}_{\mathrm{Meituan}}}$	$\frac{\mathrm{LSR}_{\mathrm{FlexDA}}}{\mathrm{LSR}_{\mathrm{Meituan}}}$
New fresh products	0.18	58%	86%
New standard products	0.10	85%	87%

In the model assembling stage, the weights of submodels are trained with the loss function $\lambda \cdot \mathrm{IR} + \mathrm{LSR}$, where λ can significantly impacts the performance metrics LSR and IR. To validate the robustness of the FlexDA approach, we conduct extensive experiments with different values of λ. In addition to Meituan's existing algorithm, we also implement two machine learning models as benchmarks: the random forest model and the XGBoost model, which train shared models with fully aggregated data sets. The average metric values of LSR and IR for a one-week test period under different values of λ are normalized due to confidentiality requirements and reported in Figure 7.

First, it's critical to recognize the tradeoff between the metrics LSR and IR. This implies that efforts to enhance one of these metrics inevitably lead to a potential loss in the other metric. Therefore, the optimization process requires careful balancing between two metrics. Second, the metrics of Meituan's existing algorithm lie below the LSR-IR curves of the random forest and XGBoost models, but above the LSR-IR curve of our proposed model, indicating that Meituan's existing algorithm outperforms the commonly used machine learning models such as XGBoost and random forest. Thus, this finding not only endorses the superior performance of the FlexDA method over Meituan's existing algorithm but also thoroughly demonstrates the robustness of the FlexDA model with different choices of parameter λ.

Figure 7. (Color online) LSR-IR Curves of Different Approaches



8. Conclusion

In this study, we propose a novel flexible data aggregation framework that allows for various levels of data aggregation and integrates the submodels trained with different aggregated data sets with a meta-model. This FlexDA method strikes the right balance between bias and variance in data-driven predictions and decisions by capturing various degrees of homogeneity across products. For demand prediction with linear models, we theoretically demonstrate the advantages of the FlexDA method over the conventional data aggregation approaches. With nonlinear demand models and general cost functions, we provide a performance guarantee on the FlexDA approach by characterizing a finite-sample high-probability bound on the optimality gap. We further develop an efficient algorithm to facilitate the implementation of the FlexDA method in practice. We conduct comprehensive numerical studies with both synthetic and real data to validate that the FlexDA approach, which demonstrates superior performance over benchmarks consistently. Finally, through an internal study with Meituan, we demonstrate the superior performance of the FlexDA method compared with the state-of-the-art benchmarks, including the one implemented at Meituan, for the ordering problems of their CGB business. For the fresh (standard) products, the FlexDA model reduced IR and LSR, the key performance metrics adopted by Meituan, by 42% (15%) and 14% (13%), respectively, compared with the existing algorithm implemented at Meituan. This improvement can lead to Meituan saving millions of dollars per year for their CGB business, which is valued at US$45.6 billion in 2023.

The FlexDA method is applied to address demand prediction and decision making in the retailing industry under a small-data large-scale regime. With its flexibility in modeling and data aggregation, the FlexDA method can be applied to a variety of business settings, such as data-driven assortment planning and pricing, and online learning for censored demand. In this work, we utilize the product characteristics and domain knowledge to decide the data aggregation levels. One may extend the framework to jointly determine the partitions of data sets and parameter estimation based on the distribution of data and contextual information.

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Endnotes

[1] One can alternatively perform regression within each leaf rather than simple averaging. In that case, the comparison between the FlexDA, the decoupled OLS estimator, and the shared OLS estimator in Section 5 provides justification for why FlexDA may still be a preferable choice even when regression is employed within each leaf.

[2] In the absence of specific criteria, we refer readers to the Online Appendix D for guidance on selecting the number of levels.

[3] Based on the definition of IR and ISR, the effective overage cost and underage cost can be treated as $(\sum_{a=1}^{K} p_a d_a)/(\sum_{a=1}^{K} d_a)\lambda$ and p_k, respectively. Therefore, the actual quantile should be $p_k/((\sum_{a=1}^{K} p_a d_a)/(\sum_{a=1}^{K} d_a)\cdot\lambda + p_k)$. In other words, if we know the realized demand, then we can just rescale λ to obtain the correct quantile. However, as the realized demand is unknown beforehand, we just stick to the original λ for construct the quantile so that the corresponding objective is approximately by $(\sum_{a=1}^{K} p_a d_a)/(\sum_{a=1}^{K} d_a)\lambda\cdot$ IR + LSR.

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